Project: Worked examples
Title: Heat sink thermal resistance chain
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Steady-state source (junction) temperature from a series chain of thermal resistances to ambient (junction to case, case to heat sink, heat sink to ambient), with each stage's temperature rise, the margin against the permitted maximum and the largest total resistance the limit allows: the heat sink budget.
$\Delta T_i = P \cdot R_i$; $R_{tot} = \Sigma R_i$; $\Delta T_{tot} = P \cdot R_{tot}$; $T_1 = T_a + \Delta T_{tot}$ and each node below it is the one above less that stage's rise. Margin = $T_{max} - T_1$; $R_{max} = (T_{max} - T_a)/P$; $u = \Delta T_{tot}/(T_{max} - T_a)$.
Ref: JEDEC JESD51-1 — Integrated circuit thermal measurement method: electrical test method (junction-to-reference resistances)
Ref: Çengel & Ghajar, Heat and Mass Transfer §15-2 — Thermal resistance networks in the cooling of electronic equipment
Ref: Incropera & DeWitt, Fundamentals of Heat and Mass Transfer §3.1 — Thermal resistance in series

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
P	Steady power dissipated (the heat flow through every stage)	40	W	spec: DS-PWR-902 device and heat sink datasheets rev A	0 … 10000 W
T_a	Ambient the last stage rejects to	40	degC	spec: SPEC-EL-05 thermal limits	-40 … 85 degC
stages	Thermal resistances in series, hottest first	3 row(s)		spec: DS-PWR-902 device and heat sink datasheets rev A	any
stages[1]		name = junction to case; R = 0.5 K/W			
stages[2]		name = case to sink; R = 0.2 K/W			
stages[3]		name = sink to ambient; R = 1.8 K/W			
T_max	Maximum permitted temperature of the hottest node (junction or case)	125	degC	spec: SPEC-EL-05 thermal limits	40 … 200 degC

3. Assumptions

A1. Steady state: the whole dissipated power flows through the one chain, and each resistance is constant. (No thermal capacitance (nothing about warm-up or a transient overload); no parallel path through the board.)
A2. The stated ambient is the temperature the last stage actually rejects to. (Air preheated by neighbouring parts can be well above room temperature.)

4. Working

[4] R_tot = ∑ R_i — Incropera §3.1
= 0.5 + 0.2 + 1.8 K/W
= 2.5 K/W
Total thermal resistance

[5] dT_tot = P * R_tot
= 40 W × 2.5 K/W
= 100 K
Total rise from ambient to the source

[6] T_1 = T_a + dT_tot — Çengel §15-2
= 40 degC + 100 K
= 140 degC
Source temperature (hot side of junction to case)

[7] dT_1 = P · R_1
= 40 W × 0.5 K/W
= 20 K
Rise across junction to case

[8] dT_2 = P · R_2
= 40 W × 0.2 K/W
= 8 K
Rise across case to sink

[9] T_2 = T_1 − dT_1
= 140 °C − 20 K
= 120 degC
Hot side of case to sink

[10] dT_3 = P · R_3
= 40 W × 1.8 K/W
= 72 K
Rise across sink to ambient

[11] T_3 = T_2 − dT_2
= 120 °C − 8 K
= 112 degC
Hot side of sink to ambient

[12] dT_m = T_max - T_1
= 125 degC − 140 degC
= -15 K
Thermal margin (negative when over the limit)

[13] u = dT_tot / (T_max - T_a)
= 100 K / (125 degC − 40 degC)
= 1.1765
Utilisation: rise over permitted rise

[14] R_max = (T_max - T_a) / P
= (125 degC − 40 degC) / 40 W
= 2.125 K/W
Largest total resistance the limit allows (heat sink budget)

5. Results

SYMBOL	VALUE	DESCRIPTION
T_1	140 degC	Source temperature
dT_m	-15 K	Thermal margin
u	1.1765	Utilisation
R_max	2.125 K/W	Heat sink budget

SYMBOL	VALUE	DESCRIPTION
R_tot	2.5 K/W	Total thermal resistance
dT_tot	100 K	Total rise
dT_1	20 K	Rise across stage 1
dT_2	8 K	Rise across stage 2
T_2	120 degC	Hot side of stage 2
dT_3	72 K	Rise across stage 3
T_3	112 degC	Hot side of stage 3

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Source temperature T_1 ≤ T_max	u = 1.1765	u ≤ 1	1.18	Çengel §15-2

7. Applicability

No warnings.

8. Verification

Template elec.thermal-chain@1.0.0 Content hash a82a7c1426781f5d Engine calc-engine/1.0.0
Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0